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Proactive Exposure hunting with Enterprise Exposure graph

Thomas Verheyden

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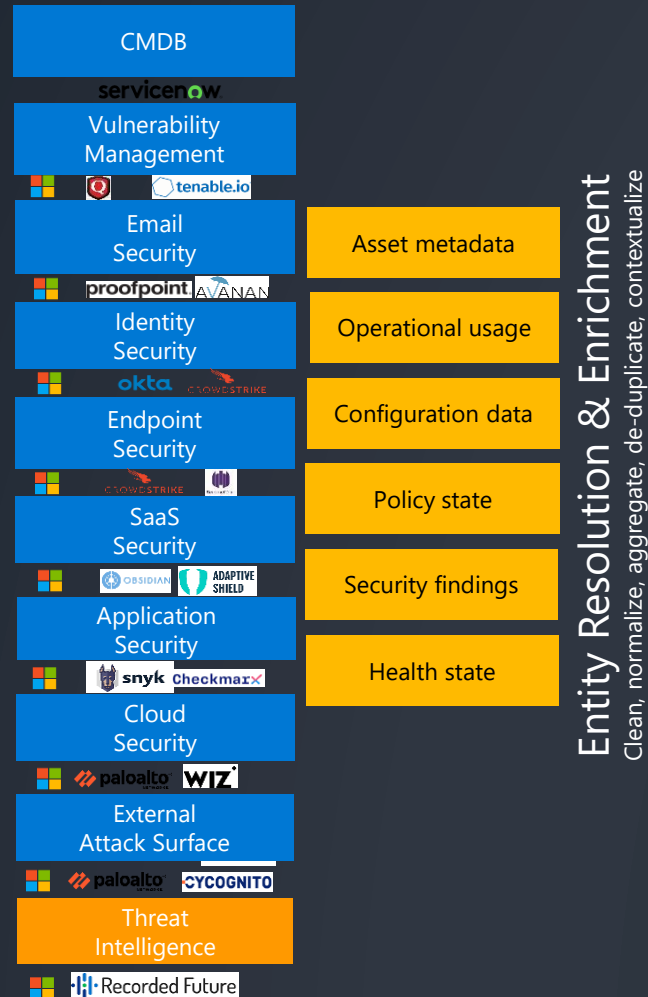




Proactive Exposure hunting with Enterprise Exposure graph

Enterprise Exposure Graph, part of Microsoft Security Exposure Management

Microsoft Security Exposure Management



Exposure Management: connectors

Default Microsoft connectors

All your connected Microsoft Security products are automatically integrated into exposure management at no extra cost. If you have the product, it's connected by default. If not, you can get it, and it will connect automatically.



Microsoft Defender For Endpoint

● Connected

Microsoft Defender for Endpoint is an enterprise endpoint security platform designed to help enterprise networks prevent, detect, investigate, and respond to advanced threats.



Microsoft Defender For Identity

● Connected

Microsoft Defender for Identity is a cloud-based security solution that helps secure your identity monitoring across your organization.



Microsoft Defender For Cloud Apps

● Connected

Microsoft Defender for Cloud Apps is a cloud access security broker (CASB) that operates on multiple clouds. It provides rich visibility, control over data travel, and sophisticated analytics to identify and combat cyberthreats across all your cloud services.



Microsoft Defender For Cloud

● Connected

Microsoft Defender for Cloud is a cloud-native application protection platform (CNAPP) that is made up of security measures and practices that are designed to protect cloud-based applications from various cyber threats and vulnerabilities.



Microsoft Defender For Office

● Connected

Microsoft Defender for Office 365 is a seamless integration into Microsoft 365 subscriptions that protects against threats in email, links (URLs), file attachments, and collaboration tools.



Microsoft Defender for Vulnerability Management

● Connected

Defender Vulnerability Management delivers asset visibility, intelligent assessments, and built-in remediation tools for Windows, macOS, Linux, Android, iOS, and network devices.



Microsoft Defender For IoT

○ Not connected

Microsoft Defender for IoT, available within the Microsoft Defender portal, is designed to secure OT environments, including networks, devices, vulnerabilities, and threats crucial for maintaining operational reliability and safety.



Microsoft Entra ID

● Connected

Microsoft Entra ID Protection helps organizations detect, investigate, and remediate identity-based risks. These identity-based risks can be further fed into tools like Conditional Access to make access decisions or fed back to a security information and event management (SIEM) tool for further investigation and correlation.

[Learn more](#)

Exposure Management: connectors

External connectors

External connectors consolidate and enrich security data across your environment, providing valuable insights



ServiceNow CMDb



Qualys



Rapid7

① Note

During the preview phase, use of the data connectors feature is free. Once data connectors become generally available, there will be a consumption-based cost for each non-Microsoft data connector based on number of assets retrieved from the connected security tool. The charge will be according to volume of ingested billable assets, where a billable asset is any asset (device, container, identity, application) on which data is reported from that connector. Each connector will have clearly defined applicable assets and guidance on how to determine the numbers. Pricing will be announced before billing of external connectors starts at GA.

Connect your Tenable service to Exposure Management to seamlessly access and manage all your devices and vulnerabilities in one place.

Connect

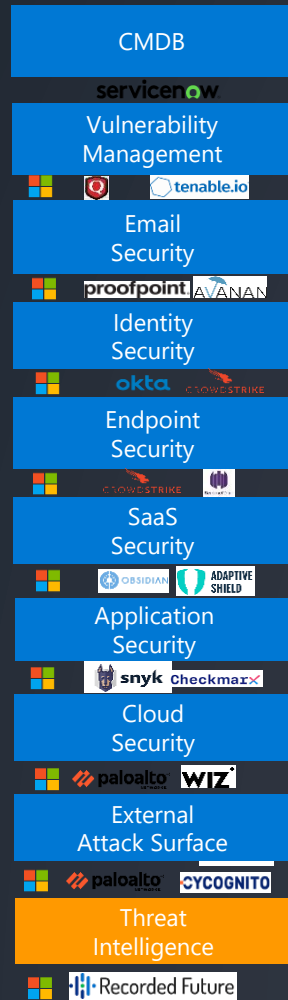
Connect your Wiz account with Microsoft exposure platform to see your devices and vulnerabilities in one place.

Connect

Connect your PaloAlto account with Microsoft exposure platform to see your devices and vulnerabilities in one place.

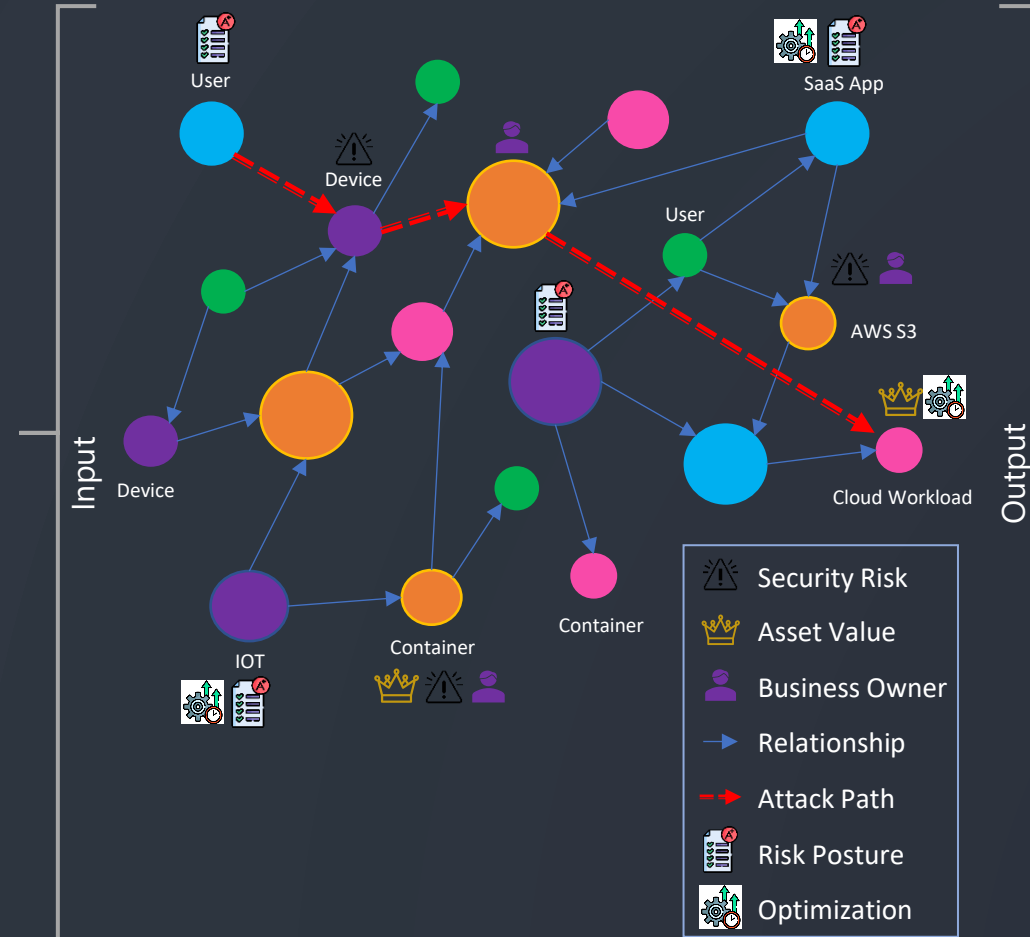
Connect

Microsoft Security Exposure Management



Entity Resolution & Enrichment

Clean, normalize, aggregate, de-duplicate, contextualize



Attack Surface Management

Continuously discover, contextualize and manage an organization's assets to provide defenders with an attacker perspective

Attack Path Analysis

Prioritize weaknesses by considering the attacker's perspective, moving away from the traditional siloed approach

Unified Exposure Insights

Understand your security posture and proactively answer questions from key stakeholders using out-of-the-box insights

Enterprise Exposure Graph

Helps security teams visualize potential attack paths, assess lateral movement risks, and prioritize remediation based on actual exposure.

Enterprise Exposure Graph

Graph structural data source

ExposureGraphNode

Contains organizational assets such as devices, users, user groups and virtual machines (VMs), along with their respective properties.

ExposureGraphEdges

Represents the relationships between the nodes in the graph. It contains edges connecting the organizational entities such as devices, identities, and virtual machines (VMs).

ExposureGraphNode

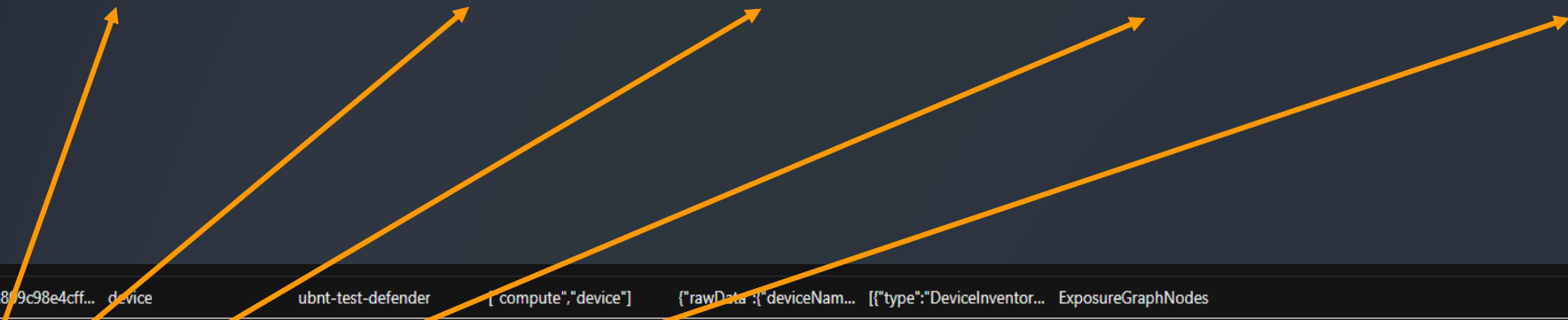
NodeID

NodeLabel

NodeName

Categories

NodeProperties



☐	50602809c98e4cff...	device	ubnt-test-defender	["compute", "device"]	{ "rawData": { "deviceName": "ubnt-test-defender", "sensorHealthState": "Active", "isAzureADJoined": false, "osVersion": "24.4", "isExcluded": false, "deviceSubtype": "Server", "firstSeenByInventory": "2024-08-20T12:47:59.2188770Z", "osPlatform": "Ubuntu" }, "type": "DeviceInventoryId", "id": "7e386b0c5e09d79244ec131871f4e5e418ff4024" }, { "type": "SenseDeviceId", "id": "7e386b0c5e09d79244ec131871f4e5e418ff4024" } }	ExposureGraphNode
NodeID	50602809c98e4cff829a2a0ee0b6a699					
NodeLabel	device					
NodeName	ubnt-test-defender					
Categories	["compute", "device"]					
NodeProperties	{ "rawData": { "deviceName": "ubnt-test-defender", "sensorHealthState": "Active", "isAzureADJoined": false, "osVersion": "24.4", "isExcluded": false, "deviceSubtype": "Server", "firstSeenByInventory": "2024-08-20T12:47:59.2188770Z", "osPlatform": "Ubuntu" }, "type": "DeviceInventoryId", "id": "7e386b0c5e09d79244ec131871f4e5e418ff4024" }, { "type": "SenseDeviceId", "id": "7e386b0c5e09d79244ec131871f4e5e418ff4024" } }					
EntityIds	[{ "type": "DeviceInventoryId", "id": "7e386b0c5e09d79244ec131871f4e5e418ff4024" }, { "type": "SenseDeviceId", "id": "7e386b0c5e09d79244ec131871f4e5e418ff4024" }]					
Type	ExposureGraphNode					

ExposureGraphNode: NodeProperties

Device

NodeProperties

```
{rawData":{"onboardingStatus":"Onboarded","osBuildRevision":"21620","machineGroup":"UnassignedGroup","osVersion":"6.3","deviceName":"prdsba...","remoteServicesInfo":{"type":"RemoteServicesInfo","wmi":{"startMode":"Auto","serviceRunning":true,"smb":{"startMode":"Auto","serviceRunning":true,"maxCvssScore":9.8,"hasHighOrCritical":true,"exploitabilityLevels":["NoExploit","ExploitsPublic"]},"rdp":{"startMode":"Auto","serviceRunning":true,"maxCvssScore":9.8,"hasHighOrCritical":true,"exploitabilityLevels":["NoExploit","ExploitsPublic"]},"winRm":{"startMode":"Auto","serviceRunning":true,"maxCvssScore":9.8,"hasHighOrCritical":true,"exploitabilityLevels":["NoExploit","ExploitsPublic"]},"rpc":{"startMode":"Auto","serviceRunning":true,"maxCvssScore":9.8,"hasHighOrCritical":true,"exploitabilityLevels":["NoExploit","ExploitsPublic"]},"rpcMap":{"startMode":"Auto","serviceRunning":true,"maxCvssScore":9.8,"hasHighOrCritical":true,"exploitabilityLevels":["NoExploit","ExploitsPublic"]},"hasAuthority":{"startMode":"Auto","serviceRunning":true,"maxCvssScore":9.8,"hasHighOrCritical":true,"exploitabilityLevels":["NoExploit","ExploitsPublic"]},"lastSeen":{"startMode":"Auto","serviceRunning":true,"maxCvssScore":9.8,"hasHighOrCritical":true,"exploitabilityLevels":["NoExploit","ExploitsPublic"]},"isCustomerFacing":{"startMode":"Auto","serviceRunning":true,"maxCvssScore":9.8,"hasHighOrCritical":true,"exploitabilityLevels":["NoExploit","ExploitsPublic"]},"isExcluded":{"startMode":"Auto","serviceRunning":true,"maxCvssScore":9.8,"hasHighOrCritical":true,"exploitabilityLevels":["NoExploit","ExploitsPublic"]}}}}
```

remoteServicesInfo

```
{"type":"RemoteServicesInfo","wmi":{"serviceRunning":true,"startMode":"Auto"},"smb":{"serviceRunning":true,"startMode":"Auto","announceServer":false,"auditSmbAccess":false,"autoShareServer":true,"autoShareWorks..."}
```

highRiskVulnerabilityInsights

```
{"type":"HighRiskVulnerabilityInsights","maxCvssScore":9.8,"hasHighOrCritical":true,"vulnerableToRemoteCodeExecution":{"maxCvssScore":9.8,"hasHighOrCritical":true,"exploitabilityLevels":["NoExploit","ExploitsPublic"]}}
```

vulnerableToRemoteCodeExecution

```
{"maxCvssScore":9.8,"hasHighOrCritical":true,"exploitabilityLevels":["NoExploit","ExploitsPublic","ExploitsVerified"]}
```

vulnerableToPrivilegeEscalation

```
{"maxCvssScore":8.8,"hasHighOrCritical":true,"exploitabilityLevels":["NoExploit","ExploitsPublic"]}
```

M365MisConfiguration

```
{category:"Security controls","activeScid":2080,"subCategory":"Credential Guard"}
```

ExposureGraphNode: NodeProperties

Entra ID users

NodeId	7601aa699262459a874d1fedf8ae0d41
NodeLabel	user
NodeName	Peter Pearson
> Categories	["environmentAzure","environmentCloud","identities","identity","user_account"]
NodeProperties	{'rawData':{'accountEnabled':true,'tags':[],'primaryProvider':'AzureActiveDirectory','aadSid':'S-1-12-1-49494829-1229849011-3935393670-3929645878','criticalityLevel':{'type':'CriticalityLevel','criticalityLevel':0,'ruleBasedCriticalityLevel':0,'ruleNames':['Compliance Administrator','Global Reader','Application Administrator','Global Administrator']}}
rawData	{'accountEnabled':true,'tags':[],'primaryProvider':'AzureActiveDirectory','aadSid':'S-1-12-1-49494829-1229849011-3935393670-3929645878','criticalityLevel':{'type':'CriticalityLevel','criticalityLevel':0,'ruleBasedCriticalityLevel':0,'ruleNames':['Compliance Administrator','Global Reader','Application Administrator','Global Administrator']}}
accountEnabled	true
tags	
primaryProvider	AzureActiveDirectory
aadSid	S-1-12-1-49494829-1229849011-3935393670-3929645878
> criticalityLevel	{'type':'CriticalityLevel','criticalityLevel':0,'ruleBasedCriticalityLevel':0,'ruleNames':['Compliance Administrator','Global Reader','Application Administrator','Global Administrator']}
accountObjectId	8c7d1818-1818-1818-1818-181818181818
accountDisplayName	Peter Pearson
accountDomain	contoso.onmicrosoft.org
accountName	Peter Pearson
accountUpn	peter.pearson@contoso.onmicrosoft.org
createdDateTime	2022-09-02T14:36:35.0000000Z
emailAddress	peter.pearson@contoso.onmicrosoft.org
firstName	Peter
hasLeakedCredentials	false
identityType	User
isActive	true
lastName	Pearson
> assignedRoles	["Global Reader","Application Administrator","Azure DevOps Administrator"]

ExposureGraphNode: NodeProperties

Hybrid User

```
"NodeProperties": {
  "rawData": {
    "primaryProvider": "Hybrid",
    "aadSid": "S-1-12-1-720320196-1193737125-1440261296-1492975556",
    "tags": [],
    "accountDomain": "domain.local",
    "accountDisplayName": "",
    "accountEnabled": true,
    "hasLeakedCredentials": false,
    "isActive": true,
    "assignedRoles": [],
    "adSid": "S-1-5-21-300511383-866583432-697575874-8243",
    "userAccountControl": [
      "NormalAccount"
    ],
    "exchangeRemoteRecipientType": 4,
    "createUsn": 23390,
    "adminCount": 0,
    "passwordUpdateTime": "2021-02-08T06:37:05.0799289Z",
    "activeDirectoryObjectId": "070ee3e5-8bfe-4f80-b0e6-593e0ef54284",
    "exchangeUserAccountControl": 0,
    "sidHistory": [],
  },
},
```

ExposureGraphNode: NodeProperties

MDC security recommendation

NodeId	59d6bb002e30417fa1cab299a7801533
NodeLabel	mdcSecurityRecommendation
NodeName	Unencrypted S3 bucket.
> Categories	["finding","security_finding"]
✓ NodeProperties	{"rawData":{"description":"Unencrypted S3 bucket.","severity":"High","securityIssue":{"type":"MdcRecommandationSecurityIssue","securityIssue":"Vulnerability"}}}
✓ rawData	{"description":"Unencrypted S3 bucket.","severity":"High","securityIssue":{"type":"MdcRecommandationSecurityIssue","securityIssue":"Vulnerability"}}}
description	Unencrypted S3 bucket.
severity	High
✓ securityIssue	{"type":"MdcRecommandationSecurityIssue","securityIssue":"Vulnerability"}
type	MdcRecommandationSecurityIssue
securityIssue	Vulnerability
> EntityIds	[{"type":"MdcAssessmentId","id":"2cd38414-1aa6-4281-bbc0-ef20c812a39d"}]
Type	ExposureGraphNode

ExposureGraphNode: NodeProperties

Cve

NodeProperties	{"rawData":{"description":"Summary: CDG through 2017-01-01 is vulnerable to directory traversal, allowing a remote attacker to download or delete arbitrary files on the system. Impact: An attacker can ex
rawData	{"description":"Summary: CDG through 2017-01-01 is vulnerable to directory traversal, allowing a remote attacker to download or delete arbitrary files on the system. Impact: An atta
description	Summary: CDG through 2017-01-01 is vulnerable to directory traversal, allowing a remote attacker to download or delete arbitrary files on the system. Impact: An attacker can ex
severity	Medium
cvssScore	5.3
cvssVectorString	CVSS:3.0/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:L/A:N/E:U/RL:U/RC:R
cvssVersion	3
exploitabilityLevel	NoExploit
exploitTypes	["Remote"]
exploitUri	
generatedTime	2025-11-04T01:00:30.4652110Z
hasExploit	false
isExploitVerified	false
isInExploitKit	false
isZeroDay	false
lastModifiedDate	2024-11-21T03:20:32.0000000Z
publishedDate	2017-01-01T00:00:00.0000000Z

ExposureGraphNode

Query

Query results are presented in your local time zone as per settings. Kusto filters, however, work in UTC. [Don't want to see it again](#)

```
1 ExposureGraphNode
2 | distinct NodeLabel
3
```

Getting started **Results** Query history

Export Show empty columns 49 items Search 00:00.824 Low Chart type Full screen

Filters: [Add filter](#)

☐	NodeLabel
☐	> entra-userCookie
☐	> device
☐	> microsoft.compute/virtualmachines
☐	> user
☐	> baseModel
☐	> microsoft.network/virtualnetworks/subnets
☐	> mdcSecurityRecommendation
☐	> microsoft.network/networkinterfaces
☐	> microsoft.storage/storageaccounts
☐	> microsoft.network/networksecuritygroups
☐	> microsoft.network/publicipaddresses
☐	> microsoft.network/routetables
☐	> network-interface-configuration
☐	> hosted-sql server
☐	> microsoft.network/virtualnetworks
☐	> Cve
☐	> group
☐	> aws-userCookie
☐	> resourcegroups
☐	> mdcManagementRecommendation
☐	> mdcAuditingRecommendation
☐	> serviceprincipal

ExposureGraphNode: NodeProperties

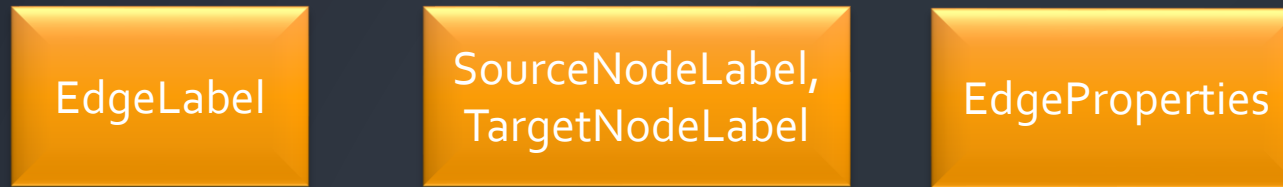
Discover internet facing devices with a privilege escalation vulnerability

^ Query

① Query results are presented in your local time zone as per settings. Kusto filters, however, work in UTC.

```
1 ExposureGraphNode
2 | where isnotnull(NodeProperties.rawData.IsInternetFacing)
3 | where isnotnull(NodeProperties.rawData.VulnerableToPrivilegeEscalation)
4 | where set_has_element(Categories, "device")
```

ExposureGraphEdges



0f0d3195-f0ac-4b...	affecting	bbed7f79a48b473089b2...	Diagnostic logs in Logic ...	mdcAuditingRecommen...	["finding","auditing_findi...	85a2ff58aabc4374ba858...	Crowdstrike_Enrichment...	microsoft.logic/workflows	["comp...
EdgeId	0f0d3195-f0ac-4b5...	bc25-06655316c30c_9593126FC09526773939A0B58F7B9C68							
EdgeLabel	affecting								
SourceNodeId	bbed7f79a48b473089b249c5198217d5								
SourceNodeName	Diagnostic logs in Logic Apps should be enabled								
SourceNodeLabel	mdcAuditingRecommendation								
SourceNodeCategories	["finding","auditing_finding"]								
TargetNodeId	85a2ff58aabc4374ba8580e...	97af7ee							
TargetNodeName	Crowdstrike_Enrichment_GetDeviceInformation								
TargetNodeLabel	microsoft.logic/workflows								
TargetNodeCategories	["compute","automation_service","environmentAzure"]								
EdgeProperties	{ "rawData":{ "publishedDate": "2025-01-16T06:52:36.9578207Z", "recommendationAzureResourceId": "/subscriptions/ef80f942-3983-4282-a93c-b6f2a66d4cab/resourcegroups/rg-sentinel-dev-01/providers/microsoft.logic/workfl								
Type	ExposureGraphEdges								

ExposureGraphEdges: EdgeProperties

Has credentials of

Query

Query results are presented in your local time zone as per settings. Kusto filters, however, work in UTC. Don't want to see it again

```
1 ExposureGraphEdges
2 | where SourceNodeLabel == "device" and TargetNodeLabel == "user" and EdgeLabel contains "has credentials of"
3 | extend Properties = parse_json(EdgeProperties)
4 | project Properties
5
```

Getting started Results Query history

Export Show empty columns 1149 items Search 00:07.119 Low Chart type Full screen

Filters: Add filter

☐ Properties

☐	>	{ "rawData": { "primaryRefreshToken": { "type": "PrimaryRefreshToken", "primaryRefreshToken": true } } }
☐	>	{ "rawData": { "primaryRefreshToken": { "type": "PrimaryRefreshToken", "primaryRefreshToken": true } } }
☐	>	{ "rawData": { "ntlmHash": { "type": "NtlmHash", "ntlmHash": true } } }
☐	>	{ "rawData": { "primaryRefreshToken": { "type": "PrimaryRefreshToken", "primaryRefreshToken": true } } }
☐	>	{ "rawData": { "primaryRefreshToken": { "type": "PrimaryRefreshToken", "primaryRefreshToken": true } } }
☐	>	{ "rawData": { "primaryRefreshToken": { "type": "PrimaryRefreshToken", "primaryRefreshToken": true } } }
☐	>	{ "rawData": { "ntlmHash": { "type": "NtlmHash", "ntlmHash": true } } }

ExposureGraphEdges: EdgeProperties

Has permission to

```
{
  "rawData": {
    "delegatedPermissions": {
      "type": "OAuthPermission",
      "permissions": [
        {
          "permissionValue": "Group.Read.All"
        },
        {
          "permissionValue": "Agreement.Read.all"
        },
        {
          "permissionValue": "Directory.ReadWrite.All"
        }
      ]
    },
    "applicationPermissions": {
      "type": "OAuthPermission",
      "permissions": [
        {
          "permissionValue": "Group.Read.All"
        }
      ]
    }
  }
}
```

Has role on

```
{
  "rawData": {
    "userRightsOnDevice": {
      "type": "RightsOnDevice",
      "userRights": [
        "SeDebugPrivilege",
        "SeImpersonatePrivilege",
        "SeBackupPrivilege",
        "SeRestorePrivilege",
        "SeLoadDriverPrivilege",
        "SeTakeOwnershipPrivilege"
      ],
      "logonMethods": [
        "remoteInteractive",
        "interactive",
        "network",
        "batch"
      ],
      "isLocalAdmin": true
    }
  }
}
```

ExposureGraphEdges: EdgeProperties

Has role on

▼	rawData	{"permissions":{"type":"Permissions","roles":[{"name":"Owner","id":"/subscriptions/2406e772-99e0-4e92-97c2-0271b9c7b6c4/providers/Microsoft.Autho
▼	permissions	{"type":"Permissions","roles":[{"name":"Owner","id":"/subscriptions/2406e772-99e0-4e92-97c2-0271b9c7b6c4/providers/Microsoft.Authorization/role
	type	Permissions
▼	roles	[{"name":"Owner","id":"/subscriptions/2406e772-99e0-4e92-97c2-0271b9c7b6c4/providers/Microsoft.Authorization/roleDefinitions/8e3af657-a8f
>	0	{"name":"Owner","id":"/subscriptions/2406e772-99e0-4e92-97c2-0271b9c7b6c4/providers/Microsoft.Authorization/roleDefinitions/8e3af657-

ExposureGraphEdges

Query

Query results are presented in your local time zone as per settings. Kusto filters, however, work in UTC. Don't want to see it again

```
1 ExposureGraphEdges
2 | distinct EdgeLabel
3
```

Getting startedResultsQuery history

Export Show empty columns

12 items Search

00:03.635 Low

Chart type

Full screen

Filters: Add filter

☐ EdgeLabel

☐ > can authenticate to

☐ > affecting

☐ > has role on

☐ > has credentials of

☐ > frequently logged ...

☐ > can authenticate as

☐ > contains

☐ > runs on

☐ > member of

☐ > has permissions to

☐ > is running

☐ > routes traffic to

Challenge

A critical Chrome Zero-Day vulnerability (CVE-2025-4664) enables attackers to steal sensitive tokens—OAuth, session IDs, and more—simply by luring users into visiting a malicious site.

```
let CriticalIdentities =
  ExposureGraphNodes
  | where set_has_element(Categories, "identity")
  | where isnotnull(NodeProperties.rawData.criticalityLevel)
  and NodeProperties.rawData.criticalityLevel.criticalityLevel < 4
  | where NodeProperties has "Global Administrator" // Remove this line to include all Critical Identities
  | distinct NodeName;
let VulnerableEndPointwithBCookie =
  ExposureGraphEdges
  | where EdgeLabel == @"has credentials of"
  | where EdgeProperties has "BrowserCookies"
  | where TargetNodeName has_any (CriticalIdentities)
  // SourceNodeName = Devices that contains GA browser cookie
  | distinct SourceNodeName;
DeviceProcessEvents
  | where Timestamp > ago(30d)
  | where ProcessVersionInfoProductName == "Google Chrome"
  | where ProcessVersionInfoProductVersion != "136.0.7103.114" and
  ProcessVersionInfoProductVersion != "136.0.7103.113"
  | summarize arg_max(Timestamp, *) by DeviceId
  | where DeviceName has_any(VulnerableEndPointwithBCookie)
```

Written by Steven Lim

[Source: Hunting-Queries-Detection-Rules/DefenderXDR/Global Admin Entra Cookie with Chrome ZeroDay.kql at main · SlimKQL/Hunting-Queries-Detection-Rules](#)

Create your own Graph structure

Make-graph operator

- **Generate** a graph structure **in memory** the output is not directly usable.

Syntax

Edges | make-graph SourceNodeId --> TargetNodeId [with_node_id= NodeIdPropertyName]

Graph-math operator

- **Define nodes and the edges between the nodes** and then gives you the ability to filter.

Create your own Graph structure

All devices with vuln and has user credentials with attribute Admincount

```
Process creations originating from PSEXEC ... X | New query* X | New query* X | New query* X | New query* X | New query* X | +
Run query Last 7 days Save Share link Create detection rule
Query
1 let Vuln = ExposureGraphNodes
2 | where (set_has_element(Categories, "device") and isnotnull(NodeProperties.rawData.highRiskVulnerabilityInsights)) or ((set_has_element(Categories, "identity") and NodeProperties.rawData.adminCount == "1"));
3 ExposureGraphEdges
4 | make-graph SourceNodeId --> TargetNodeId with Vuln on NodeId
5 | graph-match (SourceNode)-[edges]->(TargetNode)
6 | where edges.EdgeLabel == "has credentials of"
7 project edges.EdgeLabel, edges.SourceNodeName, edges.TargetNodeName, SourceNode.NodeLabel, SourceNode.NodeName, SourceNode.NodeProperties
```

Edges Source Target

☐	>	has credentials of	fa2	Libert	device	fa2	[rawData]:[osDistributi...
☐	>	has credentials of	fa1	Remi	device	fa1	[rawData]:[deviceCate...
☐	>	has credentials of	xa7	Jorde	device	xa7	[rawData]:[lastSeen]"2...
☐	>	has credentials of	fa2	Van N	device	fa2	[rawData]:[deviceCate...
☐	>	has credentials of	fa2	Depre	device	fa2	[rawData]:[deviceCate...
☐	>	has credentials of	fa1	Vand	device	fa1	[rawData]:[deviceSubt...
☐	>	has credentials of	fa2	Gavir	device	fa2	[rawData]:[osArchitect...
☐	>	has credentials of	fa2	De	device	fa2	[rawData]:[osArchitect...

Create your own Graph structure

OAuth applications with important Microsoft Graph permissions

☐ NodeOAuthApp_NodeName	NodeOAuthApp	CanAuthenticateAs_EdgeLa...	NodeServicePrincipal	HasPermissionsTo	NodeGraph_NodeName	NodeGraph_NodeLabel
SourceNodeId	bc8dbb823a0846318500d49d36b6bf49					
SourceNodeName	6db3a668-85c2-458f-9c01-60f9eec53324					

① Query results are presented in your local time zone as per settings. Kusto filters, however, work in UTC.

```
1 ExposureGraphEdges
2 | make-graph SourceNodeId --> TargetNodeId with ExposureGraphNodes on NodeId
3 | graph-match (NodeOAuthApp)-[CanAuthenticateAs]->(NodeServicePrincipal)-[HasPermissionsTo]->(NodeGraph)
4 where
5     CanAuthenticateAs.EdgeLabel =~ "can authenticate as" and
6     HasPermissionsTo.EdgeLabel =~ "has permissions to" and
7     and NodeGraph.NodeName =~ "Microsoft Graph"
8     and HasPermissionsTo.EdgeProperties has "Directory.ReadWrite.All" or HasPermissionsTo.EdgeProperties has "RoleManagement.ReadWrite.Directory"
9     or HasPermissionsTo.EdgeProperties has "AppRoleAssignment.ReadWrite.All"
10 project NodeOAuthApp.NodeName, NodeOAuthApp, CanAuthenticateAs.EdgeLabel, NodeServicePrincipal, HasPermissionsTo, NodeGraph.NodeName, NodeGraph.NodeLabel
11
```

Type	ExposureGraphEdges
SourceSystem	

Create your own Graph structure

Lets hunt for users that has any kind of role on the application which has app permissions to the MS Graph

```
ExposureGraphEdges
| make-graph SourceNodeId --> TargetNodeId with ExposureGraphNodes on NodeId
| graph-match (NodeUser)-[HasRoleOn]->(NodeOAuthApp)-[CanAuthenticateAs]->(NodeServicePrincipal)-[HasPermissionsTo]->(NodeGraph)
where
    HasRoleOn.EdgeLabel =~ "has role on" and
    CanAuthenticateAs.EdgeLabel =~ "can authenticate as" and
    HasPermissionsTo.EdgeLabel =~ "has permissions to" and
    and HasPermissionsTo.EdgeProperties has "Directory.ReadWrite.All" or HasPermissionsTo.EdgeProperties has "RoleManagement.ReadWrite.Directory"
    or HasPermissionsTo.EdgeProperties has "AppRoleAssignment.ReadWrite.All" and
    (NodeOAuthApp.NodeProperties.rawData.isBelongToExternalApplication contains "False" and NodeGraph.NodeName =~ "Microsoft Graph")
project NodeUser.NodeName, HasRoleOn, NodeOAuthApp.NodeName, NodeOAuthApp, CanAuthenticateAs.EdgeLabel, NodeServicePrincipal, HasPermissionsTo, NodeGraph.NodeName
```

Create your own Graph structure

How can an attacker obtain credentials of users? Yes, on vulnerable devices with unprotected credentials in memory. Lets add them as an additional attack path connection:

```
ExposureGraphEdges
| make-graph SourceNodeId --> TargetNodeId with ExposureGraphNodes on NodeId
| graph-match (DeviceWithRCE)-[HasCredentialsOf]->(NodeUser)-[HasRoleOn]->(NodeOAuthApp)-[CanAuthenticateAs]->(NodeServicePrincipal)-[HasPermissionsTo]->(NodeGraph)
where
    HasCredentialsOf.EdgeLabel =~ "has credentials of" and
    HasRoleOn.EdgeLabel =~ "has role on" and
    CanAuthenticateAs.EdgeLabel =~ "can authenticate as" and
    HasPermissionsTo.EdgeLabel =~ "has permissions to" and
    (NodeOAuthApp.NodeProperties.rawData.isBelongToExternalApplication contains "False" and NodeGraph.NodeName =~ "Microsoft Graph") and
    (DeviceWithRCE.Categories == ["compute","device","virtual_machine"]) and isnotempty(DeviceWithRCE.NodeProperties.rawData.VulnerableToPrivilegeEscalation)
project DeviceWithRCE.NodeName,HasCredentialsOf.EdgeLabel,NodeUser.NodeName,HasRoleOn.EdgeLabel,NodeOAuthApp.NodeName,NodeOAuthApp.NodeLabel,CanAuthenticateAs.EdgeLabel
```

Create your own Graph structure

Variable length edge

```
1 ExposureGraphEdges
2 | make-graph SourceNodeId --> TargetNodeId with ExposureGraphNodes on NodeId
3 | graph-match (n1)-[e*1..3]->(n2)
4 | where any(e, Edgelabel in ("has credentials of", "member of", "can authenticate to"))
5
```

① Semantic error

Error message

graph-match operator: variable edge 'edges', referenced in graph where clause, should be accessed using 'map', 'all' and 'any' graph functions (see <https://aka.ms/deprecation-of-variable-length-edge-dot-notation>)

How to resolve

Fix semantic errors in your query

[Source: Deprecation of variable length edge dot notation in graph-match | Microsoft Community Hub](#)

Create your own Graph structure

Zero day (CVE-2025-59232) to Domain Admins

Query

Query results are presented in your local time zone as per settings. Kusto filters, however, work in UTC. [Don't want to see it again](#)

```
1 ExposureGraphEdges
2 | make-graph SourceNodeId --> TargetNodeId with ExposureGraphNodes on NodeId
3 | graph-match (SourceNode)-[edges*2..6]->(TargetNode)
4 where TargetNode.NodeName == "Domain Admins" and SourceNode.NodeName == "CVE-2025-59232"
5 project SourceNode.NodeName, TargetNode.NodeName, edges.EdgeLabel, SourceNode.NodeLabel, SourceNode.NodeName1
6
7
```

Getting started **Results** Query history

Export Show empty columns

3828 items Search

00:47.286 Low

Chart type Full screen

Filters: Add filter

☐	SourceNode_NodeName	TargetNode_NodeName	edges_EdgeLabel	SourceNode_NodeLabel	SourceNode_NodeName1
☐	> CVE-2025-59232	Domain Admins	["affecting","has credentials of","member of"]	Cve	CVE-2025-59232
☐	> CVE-2025-59232	Domain Admins	["affecting","has credentials of","member of","has role on"]	Cve	CVE-2025-59232
☐	> CVE-2025-59232	Domain Admins	["affecting","has credentials of","member of","can authenticate to","has credentials of","member of"]	Cve	CVE-2025-59232
☐	> CVE-2025-59232	Domain Admins	["affecting","has credentials of","member of","can authenticate to","has credentials of","member of"]	Cve	CVE-2025-59232
☐	> CVE-2025-59232	Domain Admins	["affecting","has credentials of","member of","can authenticate to","has credentials of","member of"]	Cve	CVE-2025-59232
☐	> CVE-2025-59232	Domain Admins	["affecting","has credentials of","member of","can authenticate to","has credentials of","member of"]	Cve	CVE-2025-59232
☐	> CVE-2025-59232	Domain Admins	["affecting","has credentials of","member of","can authenticate to","has credentials of","member of"]	Cve	CVE-2025-59232
☐	> CVE-2025-59232	Domain Admins	["affecting","has credentials of","member of","can authenticate to","has credentials of","member of"]	Cve	CVE-2025-59232
☐	> CVE-2025-59232	Domain Admins	["affecting","has credentials of","member of","can authenticate to","has credentials of","member of"]	Cve	CVE-2025-59232
☐	> CVE-2025-59232	Domain Admins	["affecting","has credentials of","member of","has role on","member of"]	Cve	CVE-2025-59232
☐	> CVE-2025-59232	Domain Admins	["affecting","has credentials of","member of","has role on","member of","has role on"]	Cve	CVE-2025-59232
☐	> CVE-2025-59232	Domain Admins	["affecting","has credentials of","member of","can authenticate to","has credentials of","member of"]	Cve	CVE-2025-59232

Create your own Graph structure

Default output $\neq$ not very readable

Filters:					
Add filter					
☐ edges_EdgeLabel	edges_SourceNodeName	edges_TargetNodeName	SourceNode_NodeLabel	SourceNode_NodeName	SourceNode_NodeName
☐ > ["can authenticate to","has credentials of","can authenticate to"]	[Dennis J. ...]	[acc ...]	user	DOC ...	("rowData":["pri
☐ > ["can authenticate to","has credentials of","can authenticate to"]	[Van ...]	[acc ...]	user	Van ...	("rowData":["aa
☐ > ["can authenticate to","has credentials of","can authenticate to"]	[Bertrand ...]	[acc ...]	user	Bertrand ...	("rowData":["aa
☐ > ["can authenticate to","has credentials of","can authenticate to"]	[Van ...]	[acc ...]	user	Van ...	("rowData":["aa
☐ > ["can authenticate to","has credentials of","can authenticate to"]	[COM ...]	[acc ...]	user	COM ...	("rowData":["tag
☐ > ["can authenticate to","has credentials of","can authenticate to"]	[TFSS ...]	[acc ...]	user	TFSS ...	("rowData":["pri
☐ > ["can authenticate to","has credentials of","can authenticate to"]	[Dusan ...]	[acc ...]	user	Dusan ...	("rowData":["tag
☐ > ["can authenticate to","has credentials of","can authenticate to"]	[Utilit ...]	[acc ...]	user	Utilit ...	("rowData":["tag
☐ > ["can authenticate to","has credentials of","can authenticate to"]	[Noel ...]	[acc ...]	user	Noel ...	("rowData":["aa
☐ > ["can authenticate to","has credentials of","can authenticate to"]	[CAL ...]	[acc ...]	user	CAL ...	("rowData":["pri
☐ > ["can authenticate to","has credentials of","can authenticate to"]	[SINI ...]	[acc ...]	user	SINI ...	("rowData":["tag
☐ > ["can authenticate to","has credentials of","can authenticate to"]	[PRD ...]	[acc ...]	user	PRD ...	("rowData":["tag
☐ > ["can authenticate to","has credentials of","can authenticate to"]	[INT ...]	[acc ...]	user	INT ...	("rowData":["pri

Create your own Graph structure

What is map() in KQL?

The map() function is used in **graph queries** to:

- **Go through each item** (like each edge or node in a path),
- **Apply an expression** to each of them,
- **Return a new list (array)** of the results.

What is strcat_array() in KQL? (string concatenates)

strcat_array() takes a list of strings (an array) and joins them together into one long string, using a separator you choose.

Create your own Graph structure

Readable as an attack path

Query results are presented in your local time zone as per settings. Kusto filters, however, work in UTC. [Don't want to see it again](#)

```
1 ExposureGraphEdges
2 | make-graph SourceNodeId --> TargetNodeId with ExposureGraphNodes on NodeId
3 | graph-match (SourceNode)-[edges*3..4]->(TargetNode)
4 where all(edges, EdgeLabel in ("has credentials of", "member of", "has role on", "Can authenticate to")) and SourceNode.NodeLabel == "device"
5 project Source = SourceNode.NodeName, Path = strcat_array(map(edges, strcat(EdgeLabel, " -> ", TargetNodeName)), " -> "), Target = TargetNode.NodeName, EdgesLabels = edges.EdgeLabel
6
7
```

Getting started **Results** Query history

Showing the first 30,000 results for this query. To get more focused results, try adding filters or summarizations.

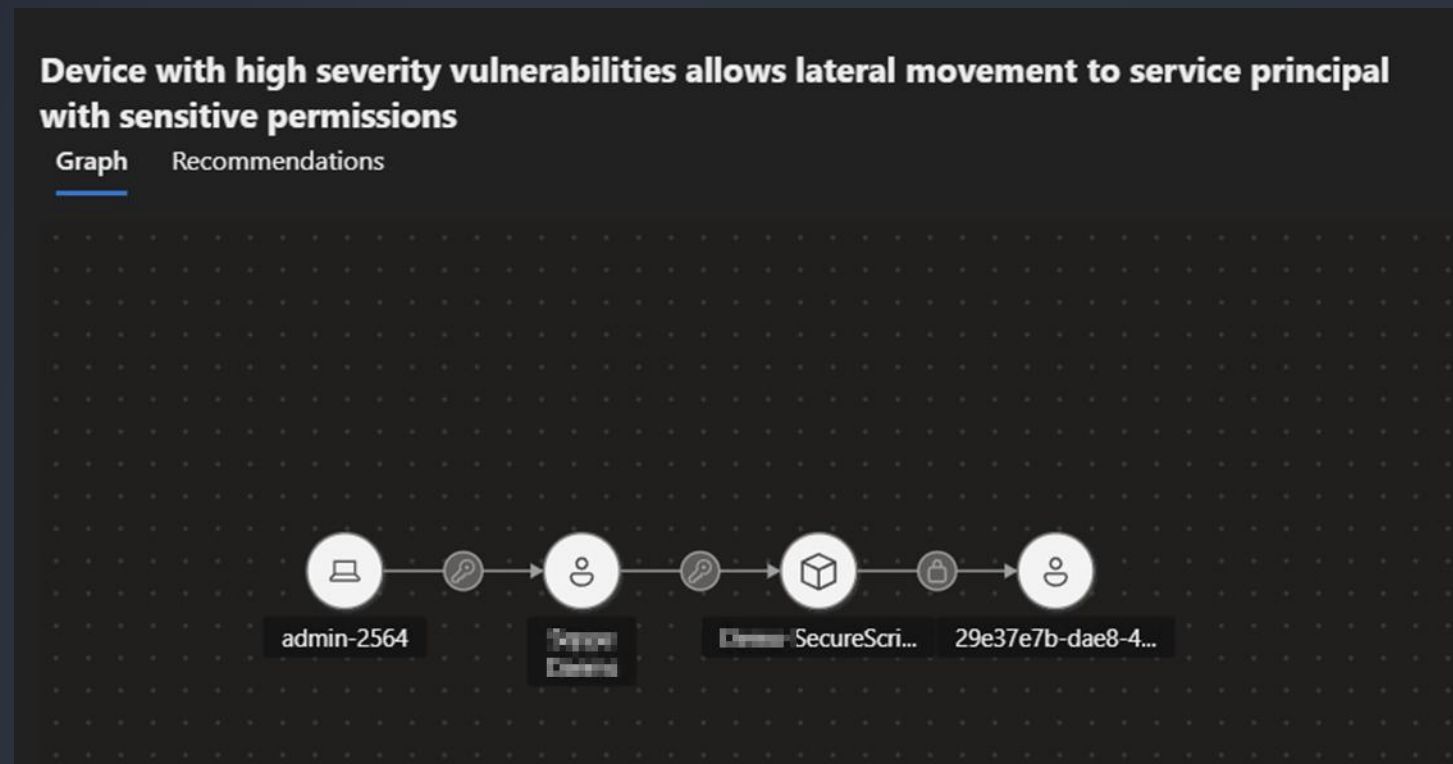
Export Show empty columns 30000 items Search 01:38.703 Low Chart type Full screen

Filters: Add filter

Source	Path	Target	EdgesLabels
> dtdag05...	has credentials of -> Ch... Cross -> member of -> GG-... -> member of -> GG-Dag...	GG-Dag...	["has credentials of", "me...
> dtdag05...	has credentials of -> Ch... Cross -> member of -> GG-... -> member of -> DL-API-...	DL-API-...	["has credentials of", "me...
> dtdag05...	has credentials of -> Ch... Cross -> member of -> GG-... -> member of -> DL-API-...	DL-API-...	["has credentials of", "me...
> dtdag05...	has credentials of -> Ch... Cross -> member of -> GG-... -> member of -> DL-M365-...	DL-M365-...	["has credentials of", "me...
> dtdag05...	has credentials of -> Ch... Cross -> member of -> GG-... -> member of -> DL-API-...	DL-API-...	["has credentials of", "me...
> dtdag05...	has credentials of -> Eli... ers -> member of -> GG-M... -> member of -> DL-M365-...	DL-M365-...	["has credentials of", "me...
> dtdag05...	has credentials of -> Eli... ers -> member of -> GG-M... -> member of -> GG-M... -> member of -> GG-M...	GG-Mob...	["has credentials of", "me...
> dtdag05...	has credentials of -> Eli... ers -> member of -> GG-M... -> member of -> DL-DIR-...	DL-DIR-...	["has credentials of", "me...
> dtdag05...	has credentials of -> Eli... ers -> member of -> GG-M... -> member of -> GG-Heelk...	GG-Heelk...	["has credentials of", "me...
> dtdag05...	has credentials of -> Eli... ers -> member of -> GG-M... -> member of -> DL-API-...	DL-API-...	["has credentials of", "me...
> dtdag05...	has credentials of -> Eli... ers -> member of -> GG-M... -> member of -> DL-DIR-...	DL-DIR-...	["has credentials of", "me...
> dtdag05...	has credentials of -> Eli... ers -> member of -> GG-M... -> member of -> GG-Inven...	GG-Inven...	["has credentials of", "me...
> dtdag05...	has credentials of -> Eli... ers -> member of -> GG-M... -> member of -> GG-Ge...	GG-Ge...	["has credentials of", "me...

Create your own Graph structure

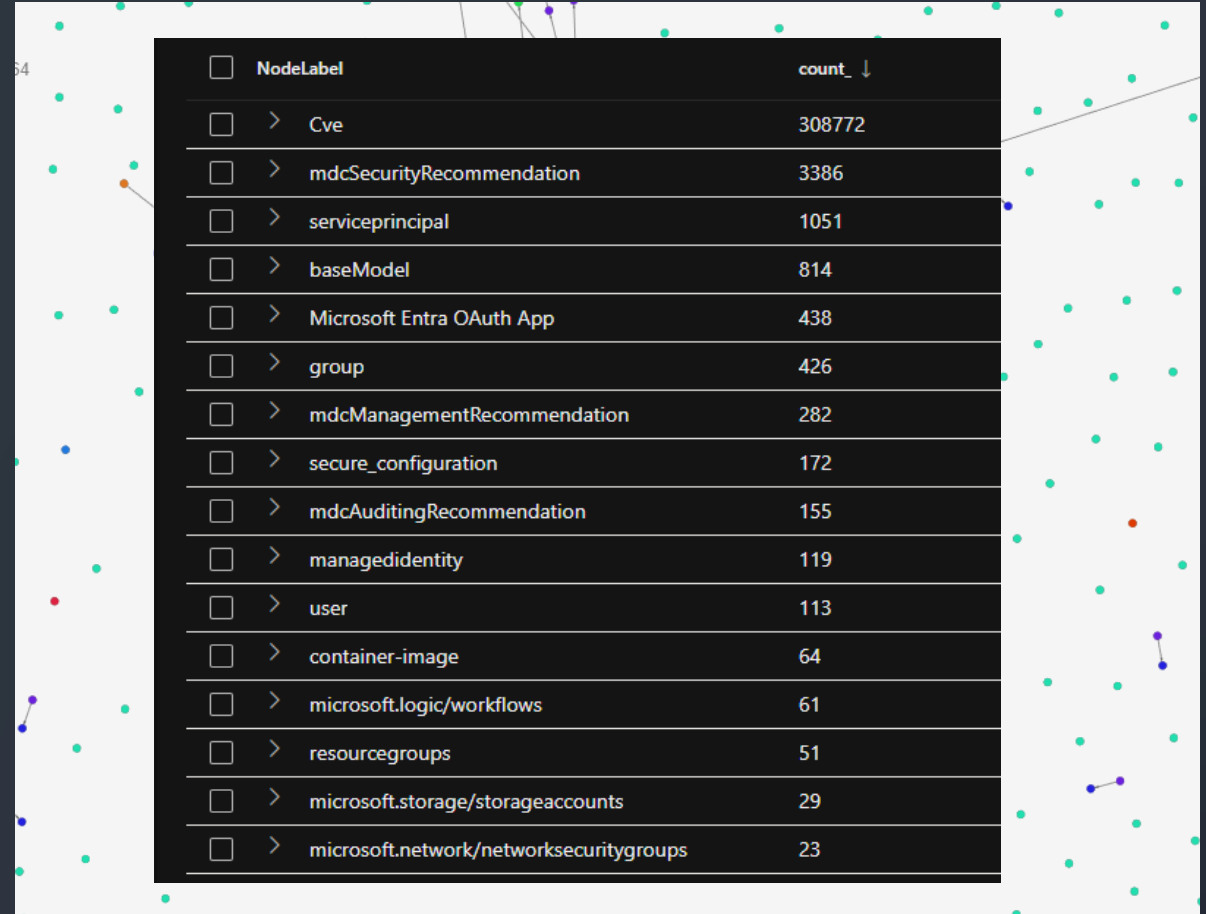
Visualization



Create your own Graph structure

Visualization

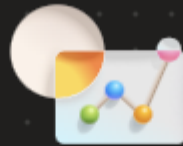
- Kusto Explorer client has now a graph visualization capabilities
- Let's use the advanced hunting API? Graph security API?
 - ➔ Limitation on 100k rows
 - ➔ Split data up based on NodeLabel
- Split data, export the data and import the data in newly created Azure Data explorer cluster
- Script available on my github: [Microsoft-Exposure-management-Kusto Explorer/AdvancedHuntingAPI . ps1 at main · v3rtho/Microsoft-Exposure-management](https://github.com/v3rtho/Microsoft-Exposure-management-Kusto-Explorer/AdvancedHuntingAPI.ps1)



☐ NodeLabel	count_ ↓
☐ > Cve	308772
☐ > mdcSecurityRecommendation	3386
☐ > serviceprincipal	1051
☐ > baseModel	814
☐ > Microsoft Entra OAuth App	438
☐ > group	426
☐ > mdcManagementRecommendation	282
☐ > secure_configuration	172
☐ > mdcAuditingRecommendation	155
☐ > managedidentity	119
☐ > user	113
☐ > container-image	64
☐ > microsoft.logic/workflows	61
☐ > resourcegroups	51
☐ > microsoft.storage/storageaccounts	29
☐ > microsoft.network/networksecuritygroups	23

Microsoft Sentinel Graph (public preview)

DEMO



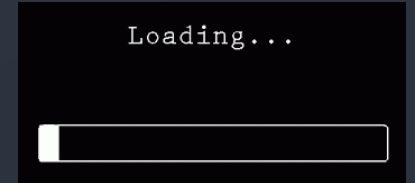
Onboard to Sentinel Graph to enable this capability

When this is complete you will be able to visually find latest threats across your organization.

[Onboard now](#)

Whats next

- Working Exposure hunting cheat sheet
 - Based on bloodhound AD & cheat sheets
- KQL queries available at:
 - Kqlsearch.com
 - v3rtho/Microsoft-Exposure-management-





The END